



Helicobacter pylori and Peptic Ulcer — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the Gram stain and shape of *Helicobacter pylori*?

EASY

- A. Gram-positive coccus
- B. Gram-negative straight bacillus
- C. Gram-negative, curved/spiral (helical) rod
- D. Gram-positive spore-forming rod
- E. Gram-negative diplococcus

2. What type of atmosphere does *H. pylori* require for growth?

EASY

- A. Strict anaerobic
- B. Obligate aerobic
- C. Capnophilic only, no oxygen tolerance
- D. Facultative anaerobic
- E. Microaerophilic

3. Which enzyme test is strongly positive for *H. pylori* and central to its acid-survival strategy?

EASY

- A. Coagulase
- B. Indole
- C. Nitrate reductase
- D. Lysine decarboxylase
- E. Urease

4. Who received the Nobel Prize for discovering *H. pylori*'s role in gastritis and peptic ulcer disease?

EASY

- A. Koch and Pasteur
- B. Fleming and Florey
- C. Watson and Crick
- D. Marshall and Warren
- E. Virchow and Metchnikoff

5. What percentage of duodenal ulcers are attributable to *H. pylori* infection?

MEDIUM

- A. About 50%
- B. About 95%
- C. About 25%
- D. About 99.9%
- E. About 10%

6. What percentage of gastric ulcers are attributable to *H. pylori* infection?

MEDIUM

- A. About 85%
- B. About 40%
- C. About 60%
- D. About 15%
- E. About 99%

7. What proportion of patients with chronic H. pylori gastritis progress to develop peptic ulcers?

MEDIUM

- A. About 1%
- B. About 50%
- C. Approximately 10–15%
- D. About 90%
- E. About 99%

8. Which virulence factor causes vacuolation and epithelial dysfunction in gastric cells?

MEDIUM

- A. CagA
- B. BabA
- C. Urease
- D. VacA (vacuolating cytotoxin A)
- E. Mucinase

9. By what mechanism does H. pylori urease neutralize the acidic gastric environment?

MEDIUM

- A. By directly pumping protons out of the bacterial cell
- B. By secreting bicarbonate ions into the mucus layer
- C. By inhibiting host parietal cell proton pumps directly
- D. Breaking down urea into ammonia and CO₂, with ammonia converted to ammonium and extruded to raise local pH
- E. By forming a protective biofilm impermeable to acid

10. Which adhesin mediates H. pylori attachment to gastric epithelial cells?

MEDIUM

- A. VacA
- B. Urease
- C. BabA (blood-antigen-binding protein A)
- D. Mucinase
- E. CagA

11. Why are CagA-positive H. pylori strains considered more virulent?

MEDIUM

- A. They lack urease activity entirely
- B. They cannot be treated with standard antibiotics
- C. They are exclusively transmitted via contaminated water
- D. They do not induce any host inflammatory response
- E. They are directly associated with acute gastritis, gastric ulcer development, and gastric carcinoma

12. A gastric ulcer characteristically develops at which anatomic site relative to inflammation?

HARD

- A. The gastroesophageal junction exclusively
- B. The duodenal bulb exclusively
- C. The junction between the corpus and antrum
- D. The pylorus exclusively, sparing the body
- E. The fundus exclusively

13. Which non-invasive test demonstrates past exposure to H. pylori but is not reliable for confirming cure?

MEDIUM

- A. Serology (antibody testing)
- B. Stool antigen test
- C. ¹³C urea breath test
- D. Biopsy-based rapid urease test
- E. Culture with susceptibility testing

14. Which factor most commonly explains treatment failure in H. pylori eradication therapy?

MEDIUM

- A. Clarithromycin resistance
- B. Patient inability to tolerate proton pump inhibitors
- C. Amoxicillin allergy in the majority of patients
- D. Spontaneous bacterial mutation to penicillin-binding protein loss
- E. Metronidazole causing treatment discontinuation in most patients

15. What is the standard duration of initial H. pylori eradication therapy?

MEDIUM

- A. 1 to 2 days
- B. 7 to 10 days
- C. 3 to 4 weeks
- D. 6 months
- E. A single dose

16. A patient completes H. pylori eradication therapy and feels completely symptom-free at follow-up. What should be done regarding confirmation of cure?

HARD

- A. No further testing is needed since symptoms have resolved
- B. Order serology only, since the patient is now asymptomatic
- C. Repeat treatment immediately as a precaution
- D. Assume cure if endoscopy shows a healed ulcer, without further microbiologic testing
- E. Perform a urea breath test, stool antigen test, or biopsy-based test at least 4 weeks after therapy, regardless of symptoms

17. Why might a rapid urease test on biopsy give a false-negative result in a patient who truly has active H. pylori infection?

HARD

- A. The rapid urease test does not actually detect urease activity
- B. Recent proton pump inhibitor or antibiotic use can suppress organism density, causing false negatives
- C. H. pylori never survives the biopsy process
- D. The test is only valid in patients under age 18
- E. Urease activity is always maximal regardless of recent treatment

18. Which sequence correctly describes the long-term pathway from H. pylori infection to gastric adenocarcinoma?

HARD

- A. Acute gastritis → immediate carcinoma, bypassing chronic changes
- B. Persistent infection → chronic active gastritis → atrophic gastritis → intestinal metaplasia/dysplasia → adenocarcinoma
- C. Atrophic gastritis → acute gastritis → adenocarcinoma directly
- D. Intestinal metaplasia → acute infection → spontaneous resolution
- E. MALT lymphoma → gastritis → peptic ulcer → carcinoma

19. Why does H. pylori preferentially colonize the gastric mucus layer near the epithelium rather than remaining free in the gastric lumen?

HARD

- A. The epithelial surface has no acid exposure at all
- B. The organism cannot survive within mucus and must attach directly to the lumen
- C. Flagella are non-functional in the mucus layer, so the organism drifts passively
- D. The pH is substantially higher near the epithelium (about 5-7) than at the luminal mucus surface (1-3), and urease plus motility allow the organism to reach this more hospitable niche
- E. Urease activity is only functional in the gastric lumen, not near the epithelium

20. A patient with longstanding untreated H. pylori infection develops a gastric mass found to be a monoclonal B-cell lymphoid proliferation. What is the most likely diagnosis and its relationship to H. pylori?

HARD

- A. MALT lymphoma, arising from chronic lymphoid infiltration of the gastric mucosa driven by persistent infection
- B. Gastric adenocarcinoma, unrelated to lymphoid tissue
- C. Burkitt lymphoma, caused by a separate viral co-infection
- D. Hodgkin lymphoma, unrelated to H. pylori status
- E. A reactive lymph node, with no malignant potential

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	H. pylori is a Gram-negative, curved, helical rod.
2	E	EASY	H. pylori is microaerophilic, requiring reduced oxygen for optimal growth.
3	E	EASY	H. pylori is a strong urease producer, which is central both to diagnosis and to acid survival.
4	D	EASY	Barry Marshall and J. Robin Warren received the 2005 Nobel Prize in Physiology or Medicine for this discovery.
5	B	MEDIUM	H. pylori is responsible for approximately 95% of duodenal ulcers.
6	A	MEDIUM	H. pylori accounts for approximately 85% of gastric ulcers.
7	C	MEDIUM	Roughly 10-15% of patients with chronic gastritis progress to peptic ulcer disease.
8	D	MEDIUM	VacA is a cytotoxin that penetrates epithelial cells and produces vacuoles, damaging epithelial function.
9	D	MEDIUM	Urease converts urea to ammonia and CO ₂ ; ammonia becomes ammonium, which is extruded and neutralizes local acid.
10	C	MEDIUM	BabA is a surface adhesin that mediates epithelial attachment.
11	E	MEDIUM	CagA-positive strains are linked to more severe disease, including acute gastritis, ulcers, and carcinoma.
12	C	HARD	Gastric ulcers commonly develop at the corpus-antrum junction, the site of intense inflammation.
13	A	MEDIUM	Serology reflects past exposure and antibody levels, which can remain positive long after successful eradication, making it unsuitable for test of cure.
14	A	MEDIUM	Treatment failure is most commonly associated with clarithromycin resistance, and susceptibility testing should follow a non-response.
15	B	MEDIUM	Standard quadruple therapy (PPI + clarithromycin + amoxicillin + metronidazole) is given for 7-10 days initially.
16	E	HARD	Symptom improvement does not prove eradication; every treated patient should have eradication confirmed using an appropriate non-serologic test, timed at least 4 weeks after therapy.
17	B	HARD	False-negative results commonly arise from recent PPI use, recent antibiotics, low organism density, or patchy colonization reducing detectable urease activity.
18	B	HARD	The Correa-type cascade proceeds from persistent infection through chronic gastritis, atrophy, and metaplasia/dysplasia before carcinoma develops.
19	D	HARD	The pH gradient across the mucus layer (acidic at the lumen, near-neutral near the epithelium) combined with urease-driven local neutralization and flagellar motility lets H. pylori reach and persist at the more hospitable epithelial surface.
20	A	HARD	Chronic H. pylori infection can drive lymphoid infiltration of the gastric mucosa, and in a small subset of patients, a monoclonal B-cell population evolves into MALT lymphoma.